

Course Description:

Introduction: An introduction to financial economics, financial assets and their roles in the economy, financial system and its management.

Choice under uncertainty and financial decisions: Definitions of uncertainty and risk, utility theory under uncertainty, axioms of choice under uncertainty, Definition of risk aversion and risk neutrality.

Financial instruments: Types and characteristics of financial instruments, term structure and theories of interest rates, i) The Expectations Hypothesis, ii) Liquidity Preference Theory, iii) Market Segmentation Theory, iv) Preferred Habitat Theory.

Options and financial derivatives: Introduction to future and option. Understanding rights and obligations of the parties involved in various types of options, Fundamental of Weiner process and Random Walk etc. Basic of Black-Scholes Pricing, Notions of Delta, Gamma, Vega, Theta , Rho. **Commodity derivatives:** price discovery, valuation of futures and options.

Investment theory and portfolio analysis: The trade-off between expected return and risk. Efficient diversification with multiple risky assets. Weak and strong form of market efficiency. The capital asset pricing model and arbitrage price model.

Financial market and economy: Nexus between Financial market and economy. A brief discussion on fragile financial system and its recurring crises. Indian economy and its relation with global financial world.

Learning Resources:

1. Z. Bodie, C. Merton and D.L. Cleeton, Financial Economics, 1st Edition, Pearson Education, 2009.
2. Stephen Roy, J. Werner and Stephen Ross, Principle of Financial Economics, 2nd Edition, Cambridge University Press, 2012.
3. J.C.Hull, Futures and Option Markets, 7th Edition, Prentice-Hall, New Jersey, 2010.
4. David G. Luenberger, Investment Science, Oxford University Press, 1997.
5. Case Studies and Research Articles.